■ 1.5.9 Limits

Here is the expression $\frac{\sin(x)}{x}$.

```
In[1]:= t = Sin[x]/x

        Sin[x]
Out[1]= ------
          x
```

If you replace x by 0, the expression becomes 0/0, and you get an indeterminate result.

```
In[2]:= t /. x->0

                                          1
Power::infy: Infinite expression - encountered.
                                          0

Infinity::indt:
      Indeterminate expresssion 0 ComplexInfinity
        encountered.

Out[2]= Indeterminate
```

If you find the numerical value of $\frac{\sin(x)}{x}$ for x close to 0, however, you get a result that is close to 1.

```
In[3]:= t /. x->0.01

Out[3]= 0.999983
```

This finds the *limit* of $\frac{\sin(x)}{x}$ as x approaches 0. The result is indeed 1.

```
In[4]:= Limit[t, x->0]

Out[4]= 1
```

Limit[*expr*, x->x_0]	the limit of *expr* as x approaches x_0

Limits.